Ansh Kumar Singh

AI Engineer

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Summary

AI Engineer delivering LLM/GenAI solutions for enterprise clients, PoC to delivery.

Experience

AI Engineer — AI & Service Transformations Jan 2025 – Present
Tata Consultancy Services (designation: Assistant Systems Engineer) *Pune, India*

- *Rapid Build Team: AI-assisted unit taking enterprise client use cases from PoC to delivery on short timelines.*
- **Built LLM document-processing PoCs end to end for three enterprise clients** *(Python, FastAPI, LangChain, Gemini, React)*:
  - **Loan & policy onboarding system** — Gemini extraction into Pydantic schemas and rule validation, with junior → senior approval and decisions emailed via Gmail API.
  - **Licence application validator** — applications cross-checked (licence, passport, address proof) against rule sets in a background job, producing a pass/fail decision and report.
  - **Conversational intake agent** — guided question flow auto-generating a Bill of Materials and session summary PDF, with prompt-level guardrails; replaced manual resource-booking request sheets.
- **Presented AI enablement sessions to executive leadership** at two enterprise clients — 4 sessions to 35–40 board members and senior executives on AI tools and agentic workflows, including a live multi-agent demo I built.

Projects

Restaurant Reservation Agent | *Python, MCP, Groq, Sarvam* | Demo video

- **Booking agent for a 75-outlet chain, no agent framework**: own tool loop, 8 MCP tools, 11-language voice.
- **Gated writes in code, not the prompt**: no booking or change until the guest confirms; full outlets reroute.
- **Built a 16-journey eval of tool choice, arguments and end state** (13/16); traced a +2 gain to 1 agent fix and 2 bugs in the eval itself.

Graph RAG for Multi-Hop QA | *Python, Pydantic, Groq, NumPy* | github.com/Ansh-66/Insurance-Graph-RAG

- **Built an end-to-end Graph RAG system over 110 scraped news articles** — relation extraction with deal status and ownership intervals, entity resolution (222 names into 136 nodes), cited answers.
- **Benchmarked against vector RAG on an 11-question set**: graph RAG answered 10/11 (vector 9/11, model alone 3/11) and 4/4 cross-document multi-hop questions vs 3/4, with every citation checked against its source.
- **11-case extraction test suite (9/11 exact)**; multi-run pass rates after output varied at temperature 0; **round-robin assembly** stopped one deal filling 14 of 30 context slots.

Retrieval-Augmented Generation Pipeline | *Python, NumPy, Gemini* | github.com/Ansh-66/rag-from-scratch

- **Diagnosed an inert LLM reranking stage, raising hit@1 from 13/17 to 17/17** — results identical to the no-rerank baseline exposed a silent fallback to cosine order; fixed the prompt contract (17 questions, 6 chunks).
- Chunking, embeddings, cosine retrieval, reranking, hit@k harness.

Skills

Languages: Python, JavaScript, SQL, HTML, CSS
AI / ML: RAG and Graph RAG (embeddings, reranking, relation extraction, entity resolution, graph traversal), LLM evaluation, LLM APIs (Gemini, OpenAI, Groq, Sarvam), MCP, tool calling, prompt engineering, agentic workflows
Frameworks & Libraries: FastAPI, SQLAlchemy, LangChain, Pydantic, NumPy, sentence-transformers, React
Tools: Git & GitHub, SQLite, Ollama

Certifications

Databricks: Generative AI Engineer Associate, ML Associate | **Anthropic**: Claude Certified Developer — Foundations

Education

The National Institute of Engineering Mysore, India
Bachelor of Engineering in Electronics & Communication *2020 – 2024*